Dreamer from Pixels: Additional Midterm Material

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Abstract

This document provides additional evidence after the original midterm draft. We keep the completed 100k-environment-step Dreamer V1 baseline as the main reference, and add two separately labeled results: a longer seed-0 run that sampled 330,732 environment steps and a teammate-reported seed-1 run at 100,000 environment steps. The longer run evaluated a checkpoint representing 330,000 environment steps and obtained 322.0840 ± 9.6868 on ten evaluation episodes. Hao Deng's seed-1 summary reports 227.0931 ± 29.4737 under the 100k protocol. The seed-1 value is useful for discussion, but its raw configuration, per-episode CSV, and checkpoint were not available locally, so we do not treat it as independently verified evidence.

1 Purpose and Scope

Our project studies Dreamer V1 for visual control in the DeepMind Control Suite. We use the fixed PyTorch implementation by Julius Frost at commit `ccea6ae4a397a94c328891bd78e81d52dd156cb6`, with `rlpyt` at commit `f04f23db1eb7b5915d88401fca67869968a07a37`. The task is `cartpole_swingup` with camera 0, 64 x 64 RGB observations, and action repeat 2. Dreamer learns a latent world model from image sequences and improves the actor and critic using imagined latent trajectories [3]. The environment and reward contract come from the DeepMind Control Suite [4].

This addendum updates the empirical part of the midterm. It does not replace the earlier report's method description, proposal-aligned future work, or Chevy Yang's contribution description. It adds Zhiyuan Yang's longer seed-0 result and records the seed-1 result supplied by Hao Deng as a provisional team comparison.

2 Protocol and Checkpoint Identity

The formal 100k protocol uses 50,000 policy decisions, corresponding to 100,000 underlying environment steps. Episodes contain 500 decisions, or 1,000 environment steps. The replay capacity is 60,000 decision transitions; training uses batches of 16 sequences of length 50, a 1,000-decision prefill, updates every 1,000 decisions, and 100 optimizer updates per update cycle. The imagination horizon is 15. The run uses 32-bit precision and `use_pcont=False`; no continuation-prediction loss is included in the world-model objective.

The longer seed-0 run used the same effective training configuration and stopped at its wall-clock limit. Its total interaction count was 165,366 decisions and 330,732 environment steps, with 16,500 optimizer updates. The reported final score belongs to `long_train_run/run_0/params.pkl`, whose payload records `itr=164999`. Thus, the evaluated checkpoint represents 165,000 decisions, or 330,000 environment steps; the additional 732 environment steps are listed separately as total

Table 1: Final Dreamer evaluation on ten episodes with environment seeds 1000–1009. The seed-1 row is a teammate-reported aggregate summary, not a locally reloaded checkpoint result.

Run	Training seed	Budget	Evidence	Mean	Sample std.
Reference baseline	0	100k env	local full artifact	181.7011	27.5705
Longer run	0	330k checkpoint	local full artifact	322.0840	9.6868
Hao Deng run	1	100k env	teammate summary	227.0931	29.4737

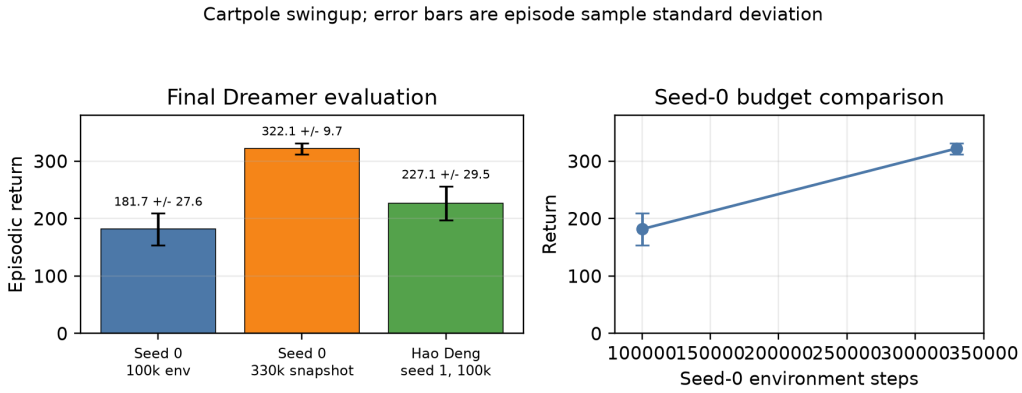


Figure 1: Additional comparison. Error bars are sample standard deviations over ten evaluation episodes. The seed-0 budget panel compares the independently evaluated 100k baseline with the longer run; it is not a learning curve. The seed-1 aggregate is shown without invented episode-level points because its raw CSV was not available locally.

sampled interaction. This distinction matters because a checkpoint counter and the runner's final interaction count do not necessarily coincide at a non-boundary stop.

The formal initial policy is the same training model saved after prefill and before the first optimizer update. We do not create a separate random initialization to serve as the training-initial comparison. Each policy is evaluated on environment seeds 1000–1009 for ten episodes. Reported error bars are sample standard deviations across evaluation episodes, not variation across training seeds. Return is an episodic control metric, not a success rate.

3 Additional Results

Table 1 lists the three relevant final-policy results. The original seed-0 100k baseline is included as a reference point. The overnight seed-0 result is a different budget and is not concatenated with the 100k curve. Hao Deng's seed-1 result is reported at the same 100k budget as the reference baseline, but the raw files were not available locally for an independent audit.

The longer seed-0 policy scores higher than its own training-initial snapshot (73.7413 ± 2.3887) and random control (45.6311 ± 37.4796). Its mean return is still only about 322 out of the theoretical episodic maximum of 1,000. The increase over the original 100k seed-0 result is consistent with additional training, but the two runs also have different wall-clock histories and checkpointing, so this is not a controlled one-factor experiment. The final episodes are not evidence of statistical significance or stable mastery across training seeds.

Hao Deng's reported seed-1 result is above the reported seed-1 training-initial value (71.9370 ± 4.8223) and random control (45.1544 ± 27.3207). It is close enough in budget and headline protocol to motivate a cross-seed comparison, but the missing raw files limit what we can claim. In particular, we cannot independently verify the exact configuration, checkpoint identity, per-episode returns, or whether every source of evaluation randomness was controlled in the same way. We therefore do not compute a three-seed mean or standard deviation from this addendum.

4 What This Evidence Supports

The seed-0 overnight run supports a narrow engineering conclusion: the selected Dreamer V1 path can train substantially beyond 100k environment steps on the Kaggle GPU route, save a model, and produce a fresh-process evaluation. The observed return increase is an empirical result for one training seed and one task. It is not a reproduction of the paper's full benchmark suite.

The teammate result supports a weaker scientific conclusion: there is an additional 100k seed-1 report that can be compared provisionally with the 100k baseline. Because the evidence is aggregate-only, it should be treated as a lead for verification rather than a finalized multi-seed result. We should not pool the 20 reported evaluation episodes as if they were 20 training seeds, and we should not use the 330k seed-0 run as a second training seed.

5 Limitations and Next Actions

The most important missing item is Hao Deng's raw seed-1 package: the effective config, progress or training log, final and pre-update checkpoints, and per-episode evaluation CSV. Once supplied, we will compare the code commit, update schedule, replay capacity, step accounting, evaluation action rule, and environment-seed handling before publishing a matched cross-seed statistic. Until then, the table and figure preserve the provenance distinction.

Our next actions are concrete. First, verify the seed-1 raw artifacts and check the 100k protocol against the repository notebook. Second, keep the seed-0 330k result as a separate longer-budget experiment rather than mixing it into the 100k baseline. Third, after the baseline evidence is aligned, run the proposal's matched-budget experiments: imagination horizons such as 5, 15, and 30, and the proposed 4-pixel random-shift augmentation. Those extensions have not been executed here, and no improvement is assumed.

Team Contributions

Chevy Yang's contribution description remains unchanged from the main report. Chevy and Zhiyuan Yang both contributed to the seed-0 Dreamer V1 baseline, including experiment setup, execution, and review of the resulting evidence. Zhiyuan Yang prepared the longer seed-0 run and this additional-material analysis. Hao Deng supplied the reported seed-1 training/evaluation summary; the detailed division of his implementation and experiment work should be confirmed against the raw files before the report is treated as final.

Repository and Sources

The project repository is `https://github.com/DDaniel-code/dreamer-pixels`. The selected implementation is software rather than a paper citation [1]; the original Dreamer paper and the PlaNet predecessor provide the research context [3, 2]. All numerical claims in this addendum are labeled by evidence level and are based on the local overnight artifacts or the teammate-provided aggregate summary.

References

[1] Julius Frost. Dreamer pytorch. GitHub repository, n.d. URL `https://github.com/juliusfrost/dreamer-pytorch`. MIT-licensed source; fixed commit ccea6ae4a397a94c328891bd78e81d52dd156cb6.

[2] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97, pages 2555–2565. PMLR, 2019. URL `https://proceedings.mlr.press/v97/hafner19a.html`.

[3] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *International Conference on Learning Representations*, 2020. URL `https://arxiv.org/abs/1912.01603`. arXiv:1912.01603, 2019 initial version.

[4] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018. URL `https://arxiv.org/abs/1801.00690`.